

Clinic Module Audit

Shifa Doctor App (web-flutter-doctor) | Document version 1.0 | June 1, 2026 | Backend: backend-kotlin-shifa

Executive summary: The clinic module is the multi-user clinic workspace inside the doctor app. It lets doctors and clinic staff operate at clinic scope — shared patient roster, doctor schedules, service catalog, treatment plans, finance, and staff invitations. It is embedded in the main shell (not a separate route or package).

1. Architecture Overview

Entry point & navigation

- **Main screen:** ClinicWorkspaceScreen — 9 inner tabs
- **Doctor shell:** Clinic tab at index 4 (DoctorShellTab.clinic)
- **Clinic staff shell:** Reduced 4-tab shell; Clinic at index 1 (primary surface)
- **Visibility:** Shown only when hasClinicWorkspaceProvider is true (≥1 clinic from GET /api/me/clinics)
- **No named route:** router.dart has no /clinic path
- **Finance nav shortcut:** Sidebar "Finance" also opens the same Clinic screen (does not auto-select Finance sub-tab)

Layer map

Layer	Location
UI	web-flutter-doctor/lib/features/clinic/
State / API	web-flutter-doctor/lib/state/clinic/
Backend controllers	ClinicWorkspace, ClinicFinance, ClinicInvitation, TreatmentPlan, PublicClinic
Auth design	backend-kotlin-shifa/docs/adr/002-clinic-staff-auth.md
Stack	Flutter + Riverpod (client); Kotlin/Spring (server)

2. File Inventory

Presentation layer

File	~Lines	Role
clinic_workspace_screen.dart	1,570	Hub + 6 inline tabs
clinic_finance_tab.dart	2,840	Finance hub, 6 sub-tabs
treatment_plan_wizard_sheet.dart	2,070	Multi-step create-plan wizard
clinic_treatment_plans_tab.dart	790	Plans ledger + PDF export
clinic_doctor_schedule_page.dart	—	Full-screen colleague booking
clinic_invitations_tab.dart	190	Receptionist invites
clinic_table_shell.dart	—	Shared DataTable shell
visit_charges_catalog_dialog.dart	80	Catalog picker (unwired)
pdf/*	—	Treatment plan PDF (en/uz)

State layer

File	Purpose
clinic_models.dart	Workspace DTOs
clinic_providers.dart	Clinics, selection, members, patients, catalog, invitations
clinic_finance_*.dart	Finance models, providers, actions
clinic_treatment_plan_*.dart	Plan models, providers, actions

Pattern: Riverpod `FutureProvider` + imperative actions. Selected clinic persisted in `SharedPreferences` per user.

3. User Personas & Access Control

Membership roles (backend)

Role	Description
OWNER	First membership per clinic; manage catalog / staff
DOCTOR	Standard clinician membership
CLINIC_ADMIN	Administrative access to roster and operations
RECEPTIONIST	Front desk — full roster and scheduling
NURSE	Limited operational access
STAFF	Generic staff role

Finance role gating (client-side)

Role	Finance tab	Manage payments
OWNER, CLINIC_ADMIN, RECEPTIONIST, DOCTOR	Visible	Yes
NURSE	Visible	No
STAFF	Visible	No

Note: `canViewFinanceProvider` exists but is not used to hide the Finance tab. Server enforces via `ClinicFinanceAccessService`.

Auth architecture (ADR 002)

- Dedicated `ClinicStaffPrincipal` + JWT role `CLINIC_STAFF`
- `ClinicMembership` binds staff to clinics
- `ClinicAccessService` centralizes clinic scope checks

4. Workspace Tabs — Capabilities

Tab 0: Overview

API: `GET /api/clinics/{id}/overview-stats`

- Clinic contact info (phone, email)
- Metrics: appointments today, active doctors, patients this month
- Quick jumps to Calendar and Patients tabs

Tab 1: Doctors

API: `GET /api/clinics/{id}/members`

- Sortable/filterable member table
- Open schedule → `ClinicDoctorScheduleRoute` (book on behalf of doctor)

Tab 2: Calendar (MVP)

- Explicit MVP hint in UI
- "My schedule" → shell Calendar tab (doctors only)
- Per-doctor buttons → colleague schedule route
- **Not** a unified clinic calendar — launcher only

Tab 3: Patients

API: `GET /api/clinics/{id}/patients?page=0&size=50`

- First page only (50 records); client-side search/sort
- Desktop: master-detail with `PatientDetailPanel`
- Mobile: list → detail navigation
- Clinic-scoped access via `clinicWorkspaceId`

Gap: No pagination UI despite backend supporting Pageable.

Tab 4: Services (Catalog)

- Unified clinic catalog + doctor profile services
- Clinic items: CRUD, activate/deactivate, doctor assignment
- Doctor-owned services: read-only (edit in Profile → Services & Pricing)

Tab 5: Treatment Plans

- Clinic-wide ledger with search, status filter, sortable columns
- Create via wizard: patient → details → doctors → services → slots → payment
- Payment modes: unpaid / full / installments
- Status changes, PDF export (en/uz)
- Also from Home quick actions and patient detail

Tab 6: Finance (6 sub-tabs)

Sub-tab	Capabilities
Dashboard	Revenue, outstanding, collection rate, overdue, top doctor earnings
By appointment	Visit ledger; view services; quick-pay
Installments	Filter, mark paid, notify patient, change status
Doctor earnings	Monthly gross/collected/outstanding per doctor
Records	Financial records CRUD via dialog
Payments	Payment history

Tab 7: Invitations

- Invite receptionist by email only (backend hardcodes RECEPTIONIST role)
- List pending/consumed invites; revoke pending
- Onboarding: invitation key → `CLINIC_STAFF` JWT

Tab 8: Settings

- Read-only: name, timezone, phone, email, address
- Clinic editing in Admin shell, not workspace

5. Backend API Surface

Workspace

Method	Path
GET	/api/me/clinics
GET	/api/clinics/{id}/overview-stats
GET	/api/clinics/{id}/members
GET	/api/clinics/{id}/patients
GET	/api/clinics/{id}/patients/{patientId}/appointments
POST	/api/clinics/{id}/appointments/{appointmentId}/visit-charges

Finance

Base: /api/clinics/{id}/finance/ — dashboard, appointment-ledger, doctor-earnings, audit, records, payments, installment-items/plans, overdue, mark-paid, notify.

Treatment plans

Base: /api/treatment-plans/ — full lifecycle, catalog items, plan-services, doctor-locations, free-slots.

Access control services

- ClinicAccessService — membership, calendar, patient roster
- ClinicFinanceAccessService — finance role checks
- VisitChargeService — visit charges → treatment plan lines

6. Cross-Module Dependencies

Module	Dependency
Shell	Tab indices, visibility rules, clinic-staff layout
Auth	CLINIC_STAFF role, receptionist registration
Patients	PatientDetailPanel, forms, documents
Calendar	CalendarScreen, ClinicDoctorScheduleRoute
Appointments	Complete flow, finance/plan invalidation
Home	Quick action → treatment plan wizard
Profile	Doctor services in Services tab (read-only)
Admin	Clinic CRUD, PDF download helper

7. What Users Can Do Today

Doctors with clinic workspace

- ✓ Switch between multiple clinics
- ✓ View overview metrics and contact info
- ✓ Browse doctors; book on colleague schedules
- ✓ Browse patients (first 50); full patient detail
- ✓ Manage service catalog
- ✓ Create/manage treatment plans (wizard, status, PDF)
- ✓ Finance operations (role-dependent)
- ✓ Invite receptionists; view/revoke invitations
- ✓ View read-only settings

Clinic staff

- ✓ Same workspace UI as primary tab
- ✓ Book on behalf of doctors
- ✓ Chat, notifications, profile in reduced shell

8. Gaps, Limitations & Tech Debt

Functional gaps

1. **Settings read-only** — no in-app clinic editing for owners
2. **Calendar MVP** — launcher only, not unified clinic calendar
3. **Patients pagination** — 50 records max in UI
4. **Finance shortcut** — does not deep-link to Finance sub-tab
5. **canViewFinanceProvider unused** — tab visible to all roles
6. **Invitations RECEPTIONIST-only** — other roles not supported in UI/API
7. **Documents tab missing** — i18n key exists; docs under Patients
8. **VisitChargesCatalogDialog unwired** — auto-derive from dental docs instead

Structural / maintainability

1. `clinic_workspace_screen.dart` monolith (~1,570 lines, 6 inline tabs)
2. `clinic_finance_tab.dart` very large (~2,840 lines)
3. `treatment_plan_wizard_sheet.dart` very large (~2,070 lines)
4. No dedicated internal routing for clinic sub-tabs
5. Backend ahead of UI (pagination, audit API, role invitations)

9. Suggested Lenses for Transformation

1. **Shell vs workspace** — one tab vs sub-app with internal routing?
2. **Tab explosion** — $9 + 6 = 15$ surfaces; group by Operations / Clinical / Financial / Admin
3. **Role-based IA** — different defaults and hidden sections per role
4. **Calendar** — biggest functional gap; MVP vs real clinic calendar
5. **Monolith split** — workspace, finance, wizard are top extraction targets
6. **Dead code cleanup** — VisitChargesCatalogDialog, unused providers, Documents i18n

Appendix: File Tree

```
web-flutter-doctor/lib/features/clinic/  
  presentation/  
    clinic_workspace_screen.dart  
    clinic_doctor_schedule_page.dart  
    clinic_treatment_plans_tab.dart  
    clinic_finance_tab.dart  
    clinic_finance_record_dialog.dart  
    clinic_invitations_tab.dart  
    clinic_table_shell.dart  
    treatment_plan_wizard_sheet.dart  
    visit_charges_catalog_dialog.dart  
  pdf/  
    treatment_plan_pdf_service.dart  
    treatment_plan_pdf_data.dart  
    treatment_plan_pdf_translations.dart  
  
web-flutter-doctor/lib/state/clinic/  
  clinic_models.dart  
  clinic_providers.dart  
  clinic_finance_models.dart  
  clinic_finance_providers.dart  
  clinic_finance_actions.dart  
  clinic_treatment_plan_models.dart  
  clinic_treatment_plan_providers.dart  
  clinic_treatment_plan_actions.dart  
  
backend-kotlin-shifa/src/main/kotlin/com/shifa/  
  web/ClinicWorkspaceController.kt  
  web/ClinicFinanceController.kt  
  web/ClinicInvitationController.kt  
  web/TreatmentPlanController.kt  
  service/ClinicAccessService.kt  
  domain/Clinic.kt, ClinicMembership.kt
```

Generated for Shifa doctor app clinic module transformation planning.